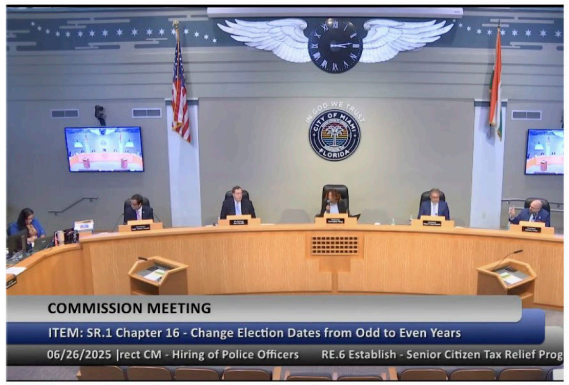
A Pipeline for Extracting Data from Videos of Complex Events

Mirya R. Holman
University of Houston

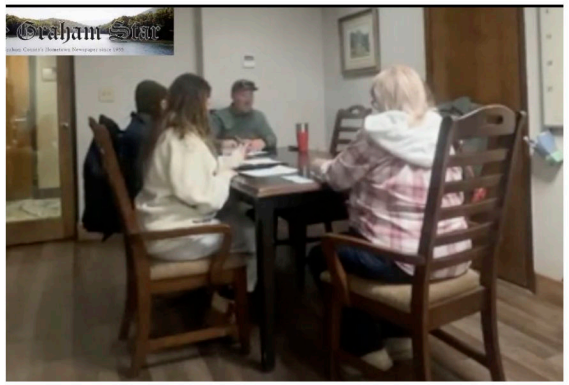
Andreas Kuepfer
TU Darmstadt

Tyler Simko
University of Michigan

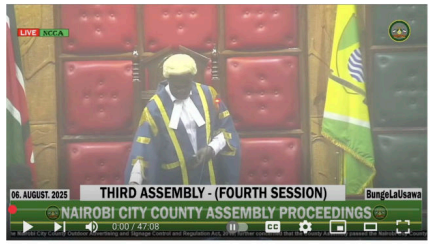
8th Annual COMPTEXT Conference (April 23-25, 2026) | University of Birmingham, UK



Miami, FL: City Commission
June 26th, 2025



Fontana Dam, NC: City Council
January 23rd, 2025



Nairobi, Kenya:
City-County Assembly

New York, NY:
City Council Candidate Forum

Lansing, MI:
Local News Interviews Mayor

Woodland Park, CO:
Council Vacancy Interviews

Mannheim, Germany:
Sitzung des Gemeinderats
(Municipal Council Meeting)

Buffalo, NY:
Local Campaign Announcement

The Research Gap in Subnational Multimodal Political Communication

- Political scientists now regularly use audio, video, and text data to study deliberation, representation, and emotion
- But: existing work focuses largely on highly professionalized settings like national legislatures

- Many institutions are forced by law to make their work transparent
- Making video recordings of political processes available is often less effort than carefully post-processing and structuring what was discussed

Studying Video Recordings of Subnational Politics

Challenges and Opportunities

Challenges and Opportunities

Subnational Policymaking is Hard to Study

- Data from subnational meetings is often not standardised
- No uniform reporting standard for meeting records, roll call votes, etc.
- Fortunately, full archives of video recordings of meetings are widely available on the institutions' websites or YouTube

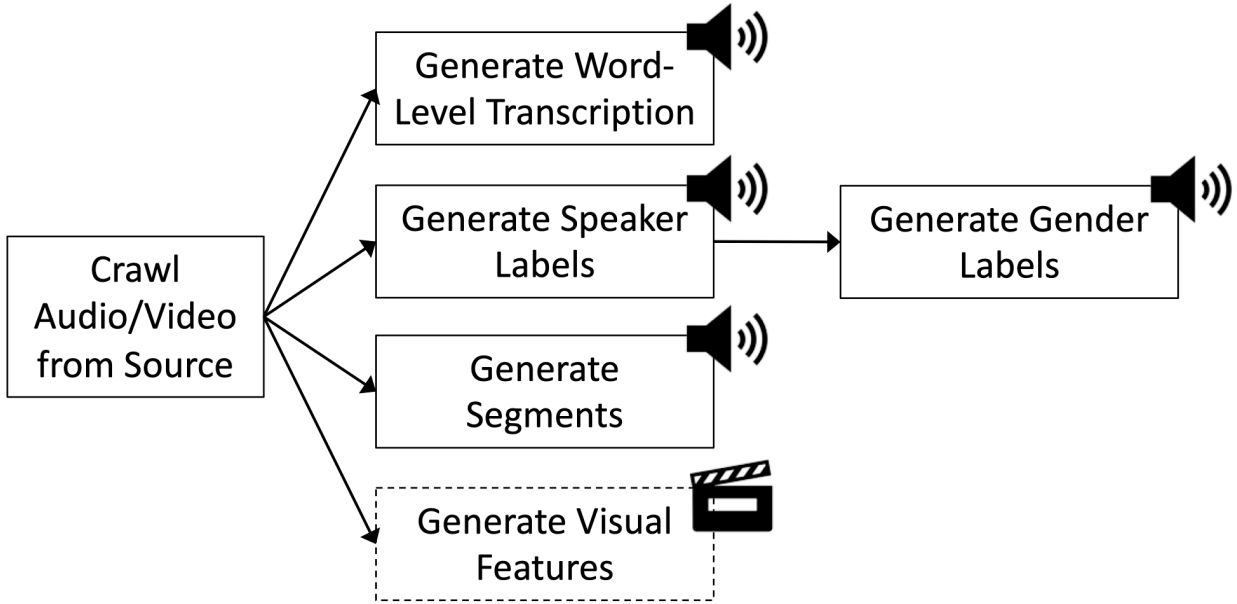
Useful Variation in Subnational Meetings

- Distributed within a country (easy to correlate with regional factors)
- Lots of over-time variation across institutions and within (e.g., school board composition, composition of justices)
- Many different actors (e.g., citizens during public comments or appellants in state supreme court meetings)

Our Study

What kinds of data can be extracted from political meetings videos, and how can these data be substantively used in political science research?

- Transcriptions
- Speaker diarization and gender classification
- Segment identification



Our Cases

US School Board Meetings

- 1,000+ hours of video (YouTube)
- 30 School Boards (564 meetings)
- Focus on variation in board gender composition

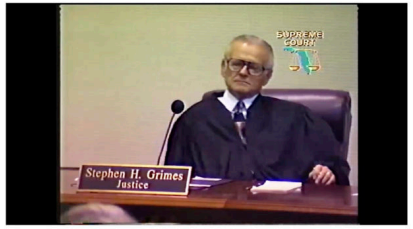
Idaho: Lakeland Joint School District 272
January 12th, 2026

Colorado: Elizabeth School District
October 9th, 2023

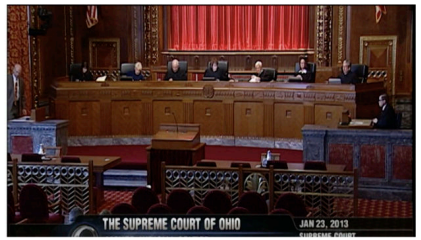
Ohio: Stow-Munroe Falls City School District
December 15th, 2025

US State Supreme Court Arguments

- 1,500+ hours of video/audio (Court websites)
- 5 State Supreme Courts
- Data spanning 1996–2025



Florida: Lecroy vs Dugger
October 9th, 1997



Ohio: Armstrong v. Jurgenson Co.
January 23rd, 2013

State of Kansas v. Justin Thurber
January 28th, 2026

The Relevance of Subnational Meetings

US School Boards

- Oversee and develop curriculum of schools
- Set local tax rates (partially)
- Approve capital projects
- Negotiations for union contracts

US State Supreme Courts

- Ultimate authority over state constitutional interpretation
- Can invalidate state laws and executive actions
- Decide high-stakes election and redistricting cases

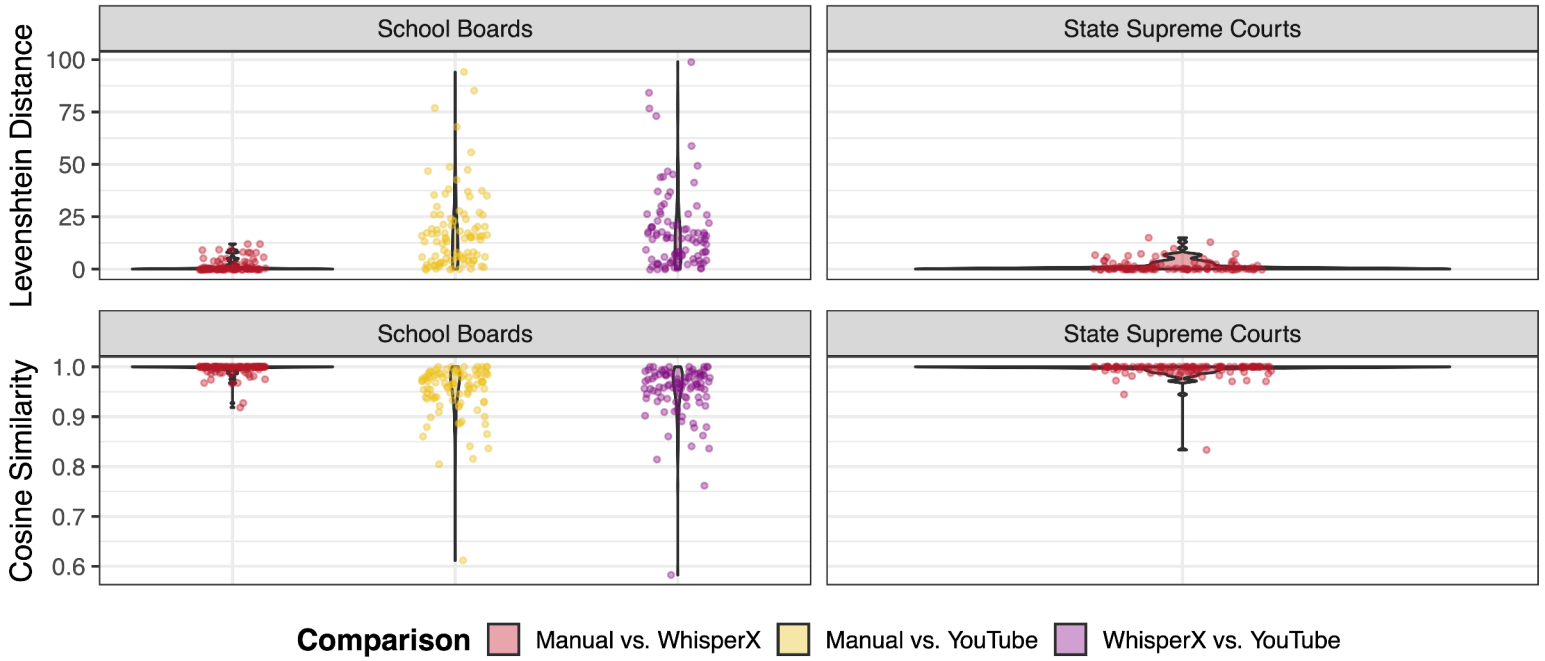
Validating the Pipeline

Transcription | Diarization | Gender | Segments

Transcription: What is Being Spoken?

Challenge: Local politics often does not produce any transcription but summarisation minutes or subtitles from YouTube.

Our Approach: *WhisperX (Bain et al. 2023)* transcribes on word-level with timestamps (1h of video = 85 seconds).



Word Error Rate: 0.009 (School Boards); 0.010 (State Supreme Courts)

Diarization: Who Speaks When?

Challenge: Local politics often does not have fixed speaker order or meta information about speakers.

Our Approach: *pyannote (Bredin 2023)* detects and aligns word-level speaker changes (1h of video = 30 seconds).

Word	Start	End	Speaker ID	Speaker Change
we	1460.49	1460.61	speaker_b061	0
have	1460.67	1460.87	speaker_b061	0
the	1460.93	1461.09	speaker_b061	0
superintendent	1461.17	1461.73	speaker_b061	0
report.	1461.77	1463.18	speaker_b061	0
Because	**1463.88**	**1464.14**	**speaker_a69c**	**1**
we	1464.18	1464.28	speaker_a69c	0
met	1464.34	1464.46	speaker_a69c	0
last	1464.52	1464.70	speaker_a69c	0

95.74% of predicted speaker changes accurate (DER: 3.79% (School Boards), 3.38% (State Supreme Courts))

What are Gendered Shares of Speech?

Challenge: Because of lacking meta information, we often do not know about gendered dynamics in local political meetings.

Our Approach: A fine-tuned *wav2vec* model assigns gender labels to audio segments (1h of video = 10 seconds).

Validation

Random sample of segments validated against manual visual coding: 98.37% correctly classified.

What Agenda Items Were Discussed and for How Long?

Challenge: Even though some institutions publish an agenda, we don't know what is actually discussed during particular agenda items.

Our Approach: Open-weight LLM *LLaMA 3.1 (8B)* generates agenda items and aligns them with start/end timestamps (1h of video = 7 seconds).

Item name	Meeting minutes	Zero shot	One shot
Front matter		0:07:23	0:07:23
Call to order	0:00:00	0:07:23	0:07:23
Pledge of allegiance	0:00:22		0:07:39
Approval of minutes	0:01:16	0:08:33	0:08:33
Communications	0:01:39	0:08:56	0:08:56
Public comment/public hearing on 2024–25 budget	**0:16:47**	**0:24:04**	**0:24:04**
Consent items	0:16:58	0:24:15	0:24:15
Action items	0:17:30	0:24:47	0:24:47
BOE planning/new business	0:55:47	1:03:17	
Adjournment	0:58:47		1:03:43

Application

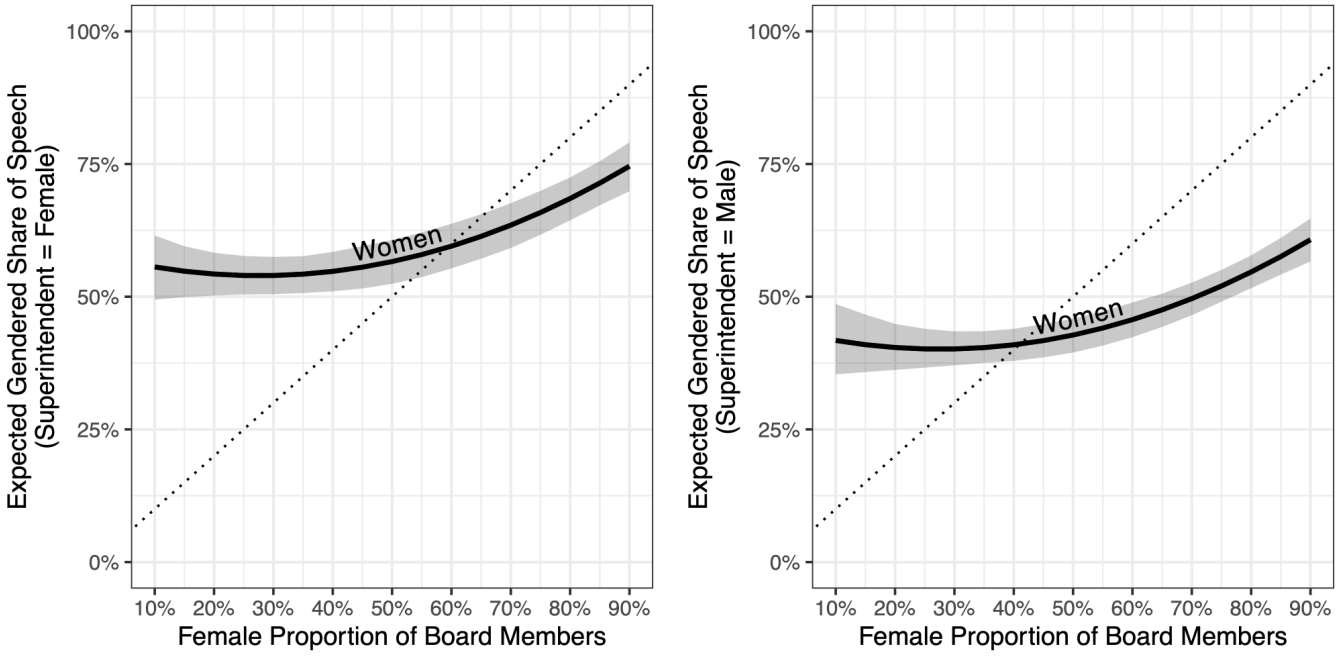
Women's Representation and Speaking Time in School Boards

Application: Women's Representation and Speaking Time

Does the gender composition of school boards translate into gendered speaking time during meetings?

- Gender of board members and superintendent hand-coded at the year level
- DV: Share of speech by women; Predictor: proportion of female board members (quadratic relationship expected)
- Separate estimates for female vs. male superintendents

Application: Key Findings



- Non-linear "critical mass" effect: Women's speech share increases sharply once female board membership exceeds ~50%
- Staff and clerks explain the floor: Even in heavily male-dominated boards, women's speech share stays ~50% which reflects administrative staff, clerks, and report authors, who are disproportionately women

Conclusion

- Subnational multimodal political communication remains limited by data access and processing challenges
- Open-source pipeline creates transcripts, diarized speech, gender labels, and segment timestamps from video data
- Dataset: 2,500+ hours of diverse video data (validated on various dimensions)
- Application: non-linear critical mass effect and staff composition shapes women's speaking time

Limitations

- Gender classifier captures *perceived* gender from vocal cues but not self-identified gender, binary only
- GPU resources required; disparities in access across institutions remain
- Scene detection unreliable for this type of video

A Pipeline for Extracting Data from Videos of Complex Events

Mirya R. Holman
University of Houston

Andreas Kuepfer
TU Darmstadt

andreas.kuepfer@tu-darmstadt.de
andreaskuepfer.github.io
ankuepfer.bsky.social

Tyler Simko
University of Michigan